CSE 351: Artificial Intelligence – Assignment 02

Student ID 1, Student ID 2

Fall 2026

Question 1: Neural Networks as Universal Approximators

Baseline Model Results

A Multi-Layer Feedforward Neural Network (MFNN) was trained using backpropagation for 5000 epochs with 1 hidden layer (40 neurons), `tansig` activation, learning rate of 0.05, and initial bias of 0.5 to approximate $f(x, y) = e^{-\frac{x^2+y^2}{8}} \cos(\sqrt{x^2 + y^2} + c)$ with $c = 1 \times 10^{-3}$.

Metric	**Value**
Initial Cost J (Epoch 0)	0.976229
Final Cost J (Epoch 5000)	0.000185
Training MSE	0.000370
Testing MSE	0.000394

Table 1: Baseline MFNN Training and Testing Results

Fine-Tuning and Hyperparameter Findings

Hyperparameter	**Configuration**	**Train MSE**	**Test MSE**
Learning Rate	lr = 0.01	0.003373	0.003497
	lr = 0.05 (Baseline)	0.000370	0.000394
	lr = 0.1	0.000116	0.000124
	lr = 0.3	0.000127	0.000150
Number of Layers	1 Layer: [40] (Baseline)	0.000370	0.000394
	2 Layers: [40, 20]	0.000051	0.000060
	3 Layers: [40, 20, 10]	0.000038	0.000049
Activation Function	tansig (Baseline)	0.000370	0.000394
	logsig	0.004096	0.004191
	relu	0.000284	0.000462
Initial Bias	bias = 0.0	0.001614	0.001686
	bias = 0.5 (Baseline)	0.000370	0.000394
	bias = 1.0	0.000063	0.000068
	bias = 2.0	0.000073	0.000090

Table 2: Effect of Hyperparameters on Train and Test MSE

Findings:

- **Number of Layers:** A single hidden layer of 40 neurons accurately approximates the function, while adding 2 and 3 hidden layers further reduces Test MSE from 0.000394 down to 0.000049 due to higher model capacity.
- **Activation Function:** `tansig` achieves the best generalization (Test MSE: 0.000394) because it is smooth and zero-centered. `logsig` converges much slower (Test MSE: 0.004191) due to smaller gradient derivatives, while `relu` produces piecewise-linear segments that slightly overfit the training points (lower Train MSE of 0.000284, but worse Test MSE of 0.000462).
- **Biases:** Setting initial bias to 0.0 gives higher error (Test MSE: 0.001686) because all hidden neurons output zero at the origin $(0, 0)$ where the target function peaks at 1.0. Non-zero biases (0.5 and 1.0) break symmetry at the origin and lower Test MSE to 0.000068, though pushing bias too high (2.0) begins to saturate the neurons.
- **Learning Rates:** A small learning rate (0.01) converges too slowly within 5000 epochs (Test MSE: 0.003497). Increasing the rate to 0.1 yields the lowest error (Test MSE: 0.000124), whereas 0.3 introduces gradient oscillations that slightly degrade final accuracy (Test MSE: 0.000150).

Question 2: Decision Tree Induction

Dataset	a_1	a_2	a_3	a_4	a_5	a_6
Data1	TBD	TBD	TBD	TBD	TBD	TBD
Data2	TBD	TBD	TBD	TBD	TBD	TBD
Data3	TBD	TBD	TBD	TBD	TBD	TBD

Table 3: GINI Index of Attributes Across Datasets

Question 3: Image Classification for Object Recognition

Classifier	**Test Accuracy (Percent)**
(a) Decision Tree	TBD
(b) Artificial Neural Network	TBD
(c) Support Vector Machine (SVM)	TBD
(d) K-Nearest Neighbour (kNN)	TBD

Table 4: CIFAR-10 Classification Accuracy